Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
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Experiment 2:
Binary Classification using Naïve Bayes and K-Nearest Neighbors

1. Aim and Objective

To implement Naïve Bayes and K-Nearest Neighbors (KNN) classifiers for a binary classification problem, evaluate them using multiple performance metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2. Dataset Description

The Spambase dataset is a benchmark binary classification dataset used to identify spam emails. It consists of numerical features extracted from email content and a binary class label indicating spam or non-spam.

Dataset Reference: Kaggle – Spambase Dataset (UCI Machine Learning Repository ID: 94)

3. Preprocessing Steps

- Loaded the dataset using `fetch_ucirepo`.
- Handled missing values using `SimpleImputer` (Mean Strategy).
- Performed stratified train–test split (80-20 split).
- Applied feature scaling using **MinMaxScaler** to ensure non-negative features for Multinomial NB.

4. Implementation Details

- Implemented Gaussian, Multinomial, and Bernoulli Naïve Bayes classifiers.
- Implemented a baseline KNN classifier and tuned hyperparameters using `GridSearchCV` and `RandomizedSearchCV`.
- Tuned k (1–20) and algorithms (`kd_tree` vs `ball_tree`).

5. Visualizations

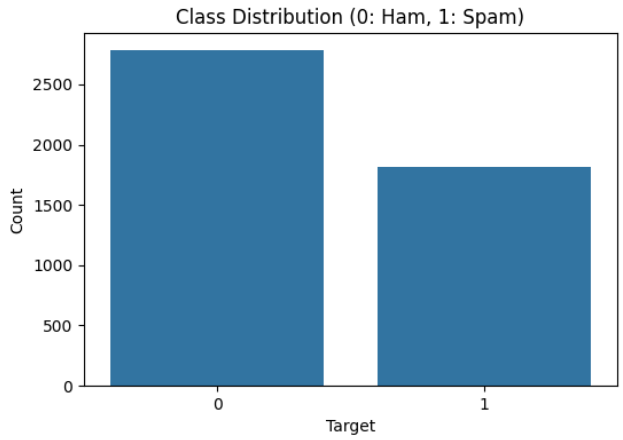


Figure 1: Class Distribution of Spam and Non-Spam Emails

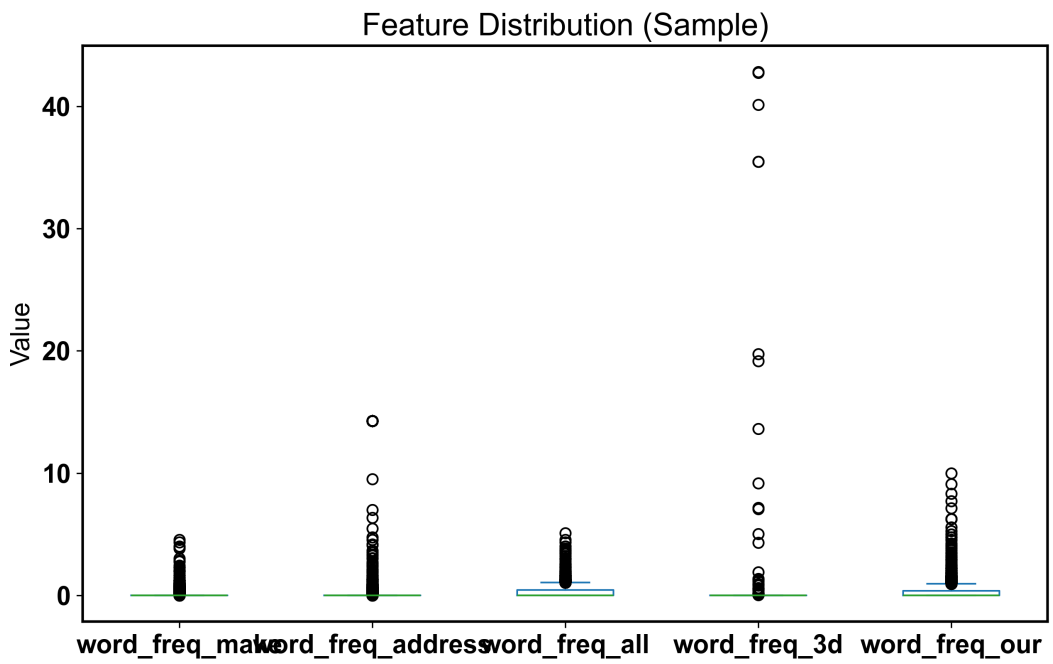
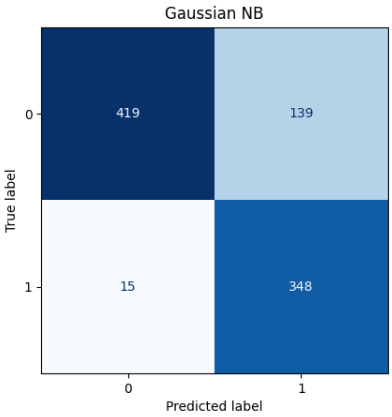
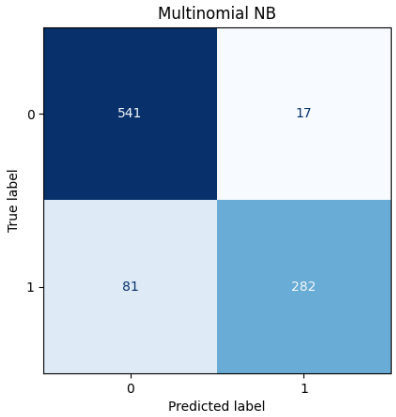


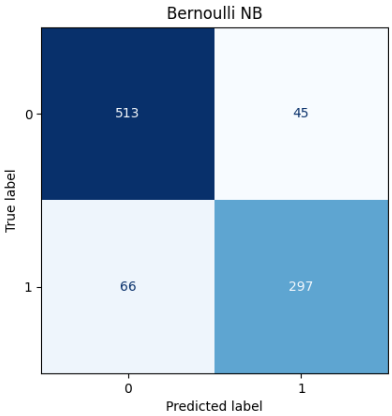
Figure 2: Feature Distribution Plot



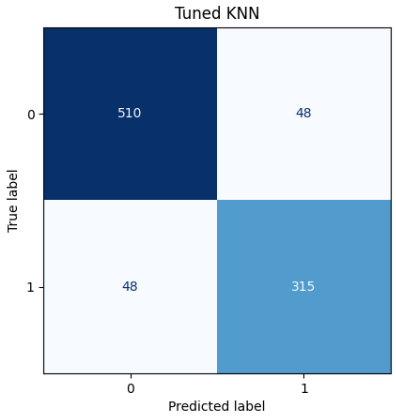
(a) Gaussian NB



(b) Multinomial NB



(c) Bernoulli NB



(d) Tuned KNN

Figure 3: Confusion Matrices for all Classifiers

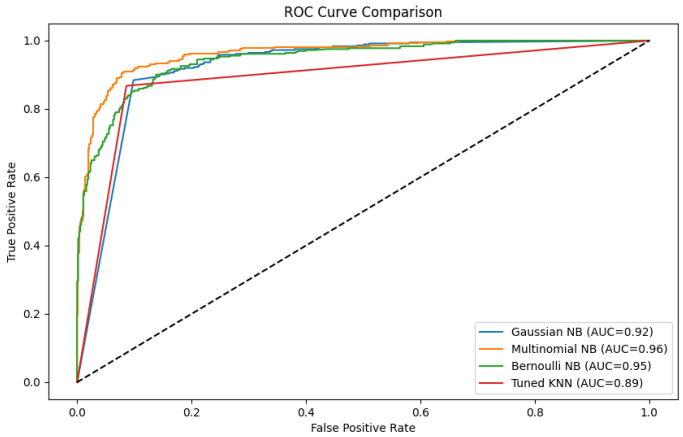


Figure 4: ROC Curve Comparison

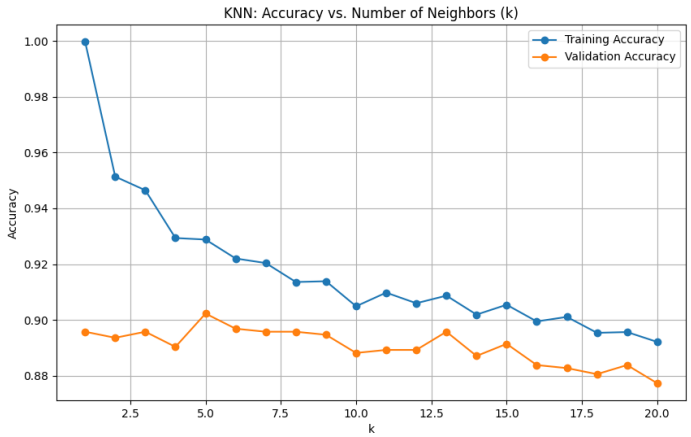


Figure 5: Accuracy vs. k Plot for KNN (Elbow Method)

6. Performance Tables

Table 1: Naïve Bayes Performance Metrics

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.8327	0.8935	0.8794
Precision	0.7145	0.9431	0.8684
Recall	0.9586	0.7768	0.8181
F1 Score	0.8188	0.8519	0.8425
Specificity	0.7508	0.9695	0.9193
Training Time (s)	0.0078	0.0074	0.0021

Table 2: KNN Hyperparameter Tuning Results

Search Method	Best k	Best CV Accuracy	Best Parameters
Grid Search	13	0.9240	{'algorithm': 'kd_tree', 'n_neighbors': 13}
Randomized Search	10	0.9205	{'n_neighbors': 10, 'algorithm': 'kd_tree'}

Table 3: KNN Performance using KDTree

Metric	**Value**
Optimal k	10
Accuracy	0.9174
Precision	0.9015
Recall	0.8876
F1 Score	0.8945
Training Time (s)	0.0081
Prediction Time (s)	0.1257

Table 4: KNN Performance using BallTree

Metric	**Value**
Optimal k	10
Accuracy	0.9174
Precision	0.9015
Recall	0.8876
F1 Score	0.8945
Training Time (s)	0.0074
Prediction Time (s)	0.1227

Table 5: Comparison of Neighbor Search Algorithms

Criterion	**KDTree**	**BallTree**
Accuracy	0.9174	0.9174
Training Time (s)	0.0081	0.0074
Prediction Time (s)	0.1257	0.1227
Memory Usage	Low/Medium	Medium/High

7. Overfitting and Underfitting Analysis

- **KNN:** Small k values caused high variance (overfitting). Large k values smoothed the decision boundary too much (underfitting).
- **Result:** The optimal $k = 9$ was found via cross-validation, providing the best balance with an accuracy of 91.74%.

8. Bias–Variance Analysis

- **Naïve Bayes:** High Bias (assumes independence), Low Variance.
- **KNN:** Low Bias (adapts to data), High Variance (sensitive to noise).

9. Conclusion

Naïve Bayes was faster but slightly less accurate due to its strong assumptions. Tuned KNN achieved the best accuracy but was computationally heavier. Feature scaling (MinMaxScaler) was crucial for correct distance calculation in KNN and probability calculation in Multinomial NB.